Überblick Zwischencode

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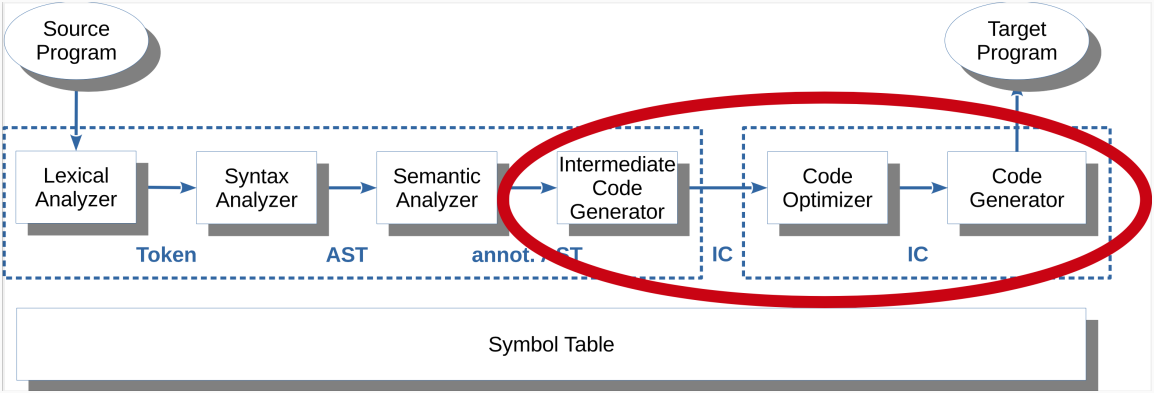
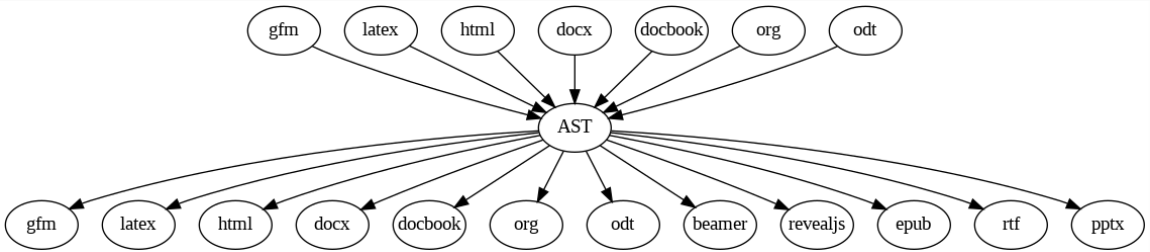
Einordnung


Source Program
Target Program
Lexical Analyzer
Syntax Analyzer
Semantic Analyzer
Intermediate Code Generator
Code Optimizer
Code Generator
Token
AST
annot. AST
IC
IC
Symbol Table

AST als Zwischencode (Beispiel Pandoc)



Konsole: pandoc hello.md -s -t native

Zwischenformat: Drei-Adressen-Code

```
i = i+1;
if (a[i] >= v) {
    i = 0;
}
```

```
      t1 = i + 1
      i  = t1
      t2 = i * 8
      t3 = a + t2
      if t3 >= v goto L1
      goto L2
L1:   i  = 0
L2:   ...
```

LLVM IR

Low **L**evel **V**irtual **M**achine

```c
int main() {
    int x = 7;
    int y = x + 35;

    return 0;
}
```

```llvm
define i32 @main() #0 {
  %1 = alloca i32, align 4
  %2 = alloca i32, align 4
  %3 = alloca i32, align 4
  store i32 0, i32* %1, align 4
  store i32 7, i32* %2, align 4
  %4 = load i32, i32* %2, align 4
  %5 = add nsw i32 %4, 35
  store i32 %5, i32* %3, align 4
  ret i32 0
}
```

Beispiel: clang -emit-llvm -S -o - hello.c

Bytecode (Beispiel Python)

```python
x = 7
y = x + 35
```

```
  1   0 LOAD_CONST     0 (7)
      3 STORE_NAME     0 (x)

  2   6 LOAD_NAME      0 (x)
      9 LOAD_CONST     1 (35)
     12 BINARY_ADD
     13 STORE_NAME     1 (y)
     16 LOAD_CONST     2 (None)
     19 RETURN_VALUE
```

Beispiel: python -m dis hello.py

Bytecode (Beispiel Java)

```java
public class Hello {

    void wuppie() {
        int x = 7;
        int y = x + 35;
    }

}
```

```
Compiled from "Hello.java"
public class Hello {
  public Hello();
    Code:
       0: aload_0
       1: invokespecial #1 // Method java/lang/Object."<init>":()V
       4: return

  void wuppie();
    Code:
       0: bipush        7
       2: istore_1
       3: iload_1
       4: bipush        35
       6: iadd
       7: istore_2
       8: return
}
```

Beispiel: javac Hello.java && javap -c Hello.class

Assembler

```c
int main() {
    int x = 7;
    int y = x + 35;

    return 0;
}
```

```asm
    .file   "hello.c"
    .text
    .globl  main
    .type   main, @function
main:
.LFB0:
    .cfi_startproc
    pushq   %rbp
    .cfi_def_cfa_offset 16
    .cfi_offset 6, -16
    movq    %rsp, %rbp
    .cfi_def_cfa_register 6
    movl    $7, -8(%rbp)
    movl    -8(%rbp), %eax
    addl    $35, %eax
    movl    %eax, -4(%rbp)
    movl    $0, %eax
    popq    %rbp
    .cfi_def_cfa 7, 8
    ret
    .cfi_endproc
.LFE0:
    .size       main, .-main
    .ident      "GCC: (Ubuntu 7.3.0-27ubuntu1~18.04) 7.3.0"
    .section    .note.GNU-stack,"",@progbits
```

Wrap-Up

- Compiler generieren aus AST Zwischencode ("*IC*" oder "*IR*")
- Kein allgemein definiertes Format, große Bandbreite:
  - AST als IR
  - LLVM IR
  - Drei-Adressen-Code
  - Diverse Arten von Bytecode
  - Assemblercode

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